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REVIEW ARTICLE

Design and Implementation of an Emergency Action Plan for Sudden Cardiac Arrest in Sport

Henry F. Pelto[1] **· Jonathan A. Drezner**[1]

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Abstract

Sudden cardiac arrest (SCA) is the leading cause of exercise-related fatalities in athletes. A comprehensive emergency action plan (EAP) is critical to facilitate a rapid and effective response to a cardiac emergency. SCA should be suspected in any athlete that collapses suddenly and is unresponsive. All potential responders to a collapsed athlete should be trained in the recognition of SCA, cardiopulmonary resuscitation, and use of an automated external defibrillator (AED). AEDs should be accessible on-site at sporting venues with a target collapse to first shock interval of less than 3 min. Every school, club, and sporting organization that sponsors athletic activities should have a written EAP for SCA. An EAP coordinator should be designated to foster compliance with training, practice, and rehearsal of the EAP at least once annually. Some sports require special considerations for equipment removal or access to emergency services in geographically broad or water-based venues.

Keywords Emergency action plan · EAP · SCA · Sudden cardiac arrest · Sudden cardiac death · Sports cardiology

Background

Sudden cardiac arrest (SCA) is the leading cause of sudden death in exercising athletes [1]. SCA in young athletes is usually associated with an underlying structural or electrical cardiac disorder, such as a cardiomyopathy or an ion channel disorder [1, 2]. Rarely, blunt trauma to the chest from a hard projectile, such as a baseball, lacrosse ball, or hockey puck, could also cause SCA in a young athlete—a condition known as commotio cordis. In athletes over the age of 35, the primary cause of SCA is atherosclerotic coronary artery disease. While SCA can occur during any sport, nearly 75% of all cases in young athletes occur in 3 sports: basketball, football, and soccer [1–5]. Male and black athletes are at higher risk [1, 3]. While the best approach to cardiovascular screening in athletes for the detection of heart conditions at risk for SCA remains an area of active research and discussion, no screening protocol will identify all conditions at risk of SCA. Thus, emergency planning for the potential occurrence of SCA during sport should be universal at all levels of play [6, 7]. This article reviews the recognition and management of SCA in athletes and the development and implementation of an emergency action plan (EAP) for SCA.

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✉ Henry F. Pelto
pelto@uw.edu

[1] University of Washington Stadium Sports Medicine Center, 3800 Montlake Blvd NE, 98195 Seattle, WA, USA

Recognition of Sudden Cardiac Arrest

Recognizing SCA is the first step towards effective management. However, most responders to a collapsed athlete will not be familiar with the signs of sudden cardiac arrest, and some features of SCA may actually distract a rescuer from proper recognition. Sports medicine professionals, coaches, and other anticipated first responders must maintain a high index of suspicion for SCA in any collapsed and unresponsive athlete. Delayed recognition of SCA can lead to critical delays or even failure to activate the emergency medical services (EMS) system, initiate cardiopulmonary resuscitation (CPR), and provide defibrillation. The following features of SCA can be misinterpreted and delay recognition [8, 9]:

1. Athletes with SCA usually collapse without warning. In some cases, athletes may momentarily stop and lean forward or put their hands on their knees before falling to the ground.
2. Athletes with SCA can fall into any position but often have their eyes open and rolled back.
3. Brief seizure-like activity, such as myoclonic arm and leg movements, shaking or quivering, occurs in over 50% of athletes with SCA. Thus, resuscitation is delayed if SCA is mistaken for a seizure.
4. Continued rapid respiratory movements immediately after collapse, or the onset of occasional gasping or agonal respirations that appear like periodic chest and abdominal movements within the first minute after SCA can be misinterpreted as normal breathing.
5. Inaccurate rescuer assessment of a pulse can also delay recognition and treatment.

SCA can occur during exertion, at the transition from intense exertion to lower levels of exertion such as walking after a short sprint, or at a pause or timeout in the action [10]. Thus, SCA should be assumed in any athlete with a sudden collapse who is unresponsive to verbal stimuli, and the EAP activated. If the mechanism of collapse in an athlete is not witnessed and there is concern for head or neck trauma, increased attention should be given to cervical spine stabilization [11]. However, efforts to protect and maintain stability of the cervical spine in a collapsed athlete are a secondary priority if the athlete is in SCA.

SCA in an athlete is typically caused by a ventricular arrhythmia, either pulseless ventricular tachycardia (VT) or ventricular fibrillation (VF), which are usually fatal if left untreated. In VF, the heart quivers and does not effectively pump blood from its chambers. The strongest predictor of survival after SCA is a short time interval from collapse to defibrillation (or first shock) [12]. In fact, survival from VF arrest declines approximately 10% every minute that defibrillation is delayed [13, 14].

Management of Sudden Cardiac Arrest

SCA can be effectively treated through the following steps: Early recognition and activation of the local EMS system, early CPR, early defibrillation, and early advanced life support and cardiovascular care in the hospital setting. Public access to automated external defibrillators (AEDs) has greatly improved survival from out-of-hospital SCA [15–18], and on-site AEDs should be a universal requirement at all sporting venues. If a player collapses and is unresponsive to verbal stimuli, rescuers should activate the EMS system, begin chest compressions (hands-only CPR), and retrieve and apply an AED as soon as possible (Fig. 1).

Recent research demonstrates that having AEDs on-site at sporting facilities and in schools to treat SCA increases survival to over 80% [19]. A 2019 investigation of mobile responders equipped with AEDs during road races in Japan reported 100% survival in 28 runners with witnessed SCA in which the mean time to CPR was 0.8 min and the mean time to AED shock was 2.2 min [20]. AEDs are safe to use and come with simple instructions including voice and visual prompts that can be followed by any trained or untrained responder. The AED will not provide a shock to the heart unless VT or VF is detected.

When responding to a collapsed athlete, the jersey or shirt should be cut and the chest exposed and dried. In a witnessed collapse, oxygenated blood is still present in the body and rescuers can apply hands-only CPR while an AED is retrieved. If a single rescuer is present, EMS should be alerted first, followed by retrieval and placement of an AED, if readily available, then initiation of CPR. If multiple rescuers are present, CPR should be initiated immediately by one rescuer while the other rescuer(s) alert EMS and retrieve the AED.

Hands-only CPR involves chest compressions at 100 compressions per minute and 2 in. (5 cm) in depth. If rescue breaths are added during the resuscitation, 2 breaths are provided every 30 chest compressions [13, 14]. All efforts should be made to minimize interruptions in chest compressions both before and after defibrillation. Chest compressions are temporarily stopped while the AED analyzes the heart rhythm. If the AED recommends a shock, rescuers should move away from touching the victim. Many AEDs are fully automated and will deliver a shock without the rescuer pressing a button. Other AEDs require the rescuer to press the shock button. Once a shock is delivered, chest compressions should be re-started immediately. This helps circulate blood to the heart muscle as it begins to contract and decreases the likelihood of recurrent VF. CPR should continue until the victim becomes responsive. The AED will re-analyze the heart rhythm every 2 min if the resuscitation continues. Delivery of appropriate chest compressions can be physically exhausting. Any individuals with CPR training within close proximity should be recruited to aid in the delivery of chest compressions (ideally rotating every 2 min during AED analysis). If available, oxygen can be applied if the athlete starts to breath on their own. Once EMS arrives, rescuers can assist in the transfer of care and more advanced life-support as needed [13, 14].

Development on an Emergency Action Plan

Every school, club, and organization that sponsors athletic activities should be prepared to respond to a collapsed athlete experiencing a cardiac emergency. An EAP for SCA with written policies and procedures is critical to ensure an efficient and structured response to a collapsed athlete (Table 1). [7,

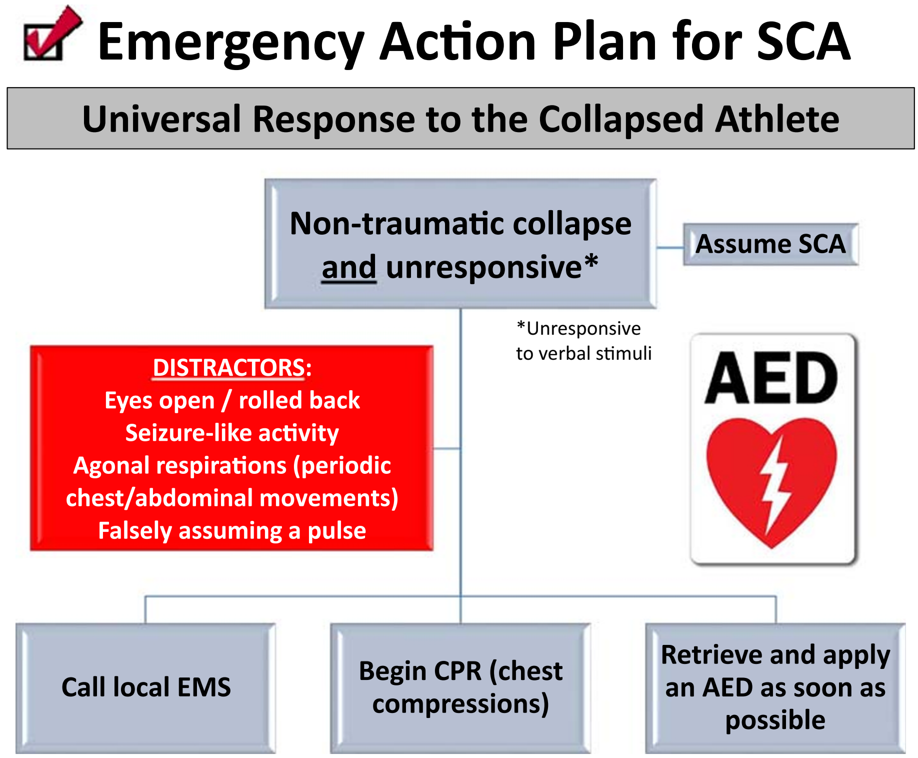

Fig. 1 Universal response to a collapsed and unresponsive athlete. Rescuers should be familiar with the potential distractors from promptly recognizing SCA. AED = automated external defibrillator; CPR = cardiopulmonary resuscitation; EMS = emergency medical services; SCA = sudden cardiac arrest

21–23] Designation of an EAP coordinator is strongly recommended to facilitate training, practice, and compliance with the EAP.

All sport medical staff and other potential first responders (i.e., coaches, facility administrators, and designated event staff) must be familiar with the EAP and trained in CPR and the use of an AED. This training should include a review of the signs of SCA; an established communication system for prompt initiation of the EAP; hands-on CPR practice for appropriate depth and rate of chest compressions; a review of the specific locations of AEDs at their practice and competition venues; and familiarity with the voice prompts and functions of the AEDs placed at their sporting venue.

Venue-Specific Emergency Action Plan

EAPs should be developed for every individual sporting venue or event site. This should include a review of the physical

Table 1 Essential elements of an emergency action plan for sudden cardiac arrest

EAP coordinator	Each school, club, or sporting organization must identify an individual responsible for coordinating their EAP.
Immediate AED access	AEDs should be placed to allow retrieval and use within 3 min of collapse at all athletic venues, fields and buildings.
Personnel training	Potential first responders to SCA including coaches must be trained in CPR and use of an AED, and organizations should maintain a roster of trained staff.
Venue-specific EAP	Each athletic venue should have its own site-specific EAP which clearly demarks the location of AEDs and transportation routes for EMS.
Communication	A reliable communication system should be established to activate EMS and alert an on-site response team if a cardiac emergency occurs.
Review and rehearsal	An annual practice drill for SCA among anticipated first responders (i.e., certified athletic trainers, physiotherapists, school nurses, coaches, and administrators) should occur prior to the start of the season.
Medical timeout	A medical timeout between home and visiting medical staff, designated coaches, and officials is recommended prior to the start of a game or competition to review the site-specific EAP.
Maintenance	AED devices must be maintained according to manufacture guidelines, including monthly readiness checks and scheduled battery and/or lead replacement.

AED automated external defibrillator, *CPR* cardiopulmonary resuscitation, *EAP* emergency action plan, *EMS* emergency medical services layout of the entire facility, familiarity with the surrounding environment (streets, buildings, barriers), and the address and directions to provide EMS to access the facility in case of an emergency. Staff members should be designated to meet arriving EMS personnel or to remove parking barriers to improve access to the venue or athletic field. This information along with the address of the facility and contact information for key personnel should be printed and displayed at the venue.

The means of communication between on-site responders and the EMS should also be established and rehearsed. This should involve identification of who will contact EMS and language to be used in communication. If the school or facility has a designated on-site response team to a medical emergency, the methods for alerting the response team should be clear and practiced.

Access to AEDs

AED location is a critical component of a venue-specific EAP. The AED should be easily and quickly accessible from the primary area of sport participation. Shock delivery as soon as possible after collapse will increase survival, and AEDs should be strategically located to allow a collapse to first shock time of less than 3 min [23]. The AED should be readily identifiable with clear signage, freely accessible, and never placed in a locked cabinet. In resource-limited environments, a single centralized AED (i.e., school gymnasium) may need to service multiple sporting venues. If an AED is not available on-site at a specific venue, the location and retrieval of the nearest AED should be discussed and rehearsed.

Transition of Care

The care of the collapsed athlete after the arrival of emergency medical personnel should also be reviewed. This should include the role of first responders once EMS takes over management. The names and locations of the closest advanced cardiac care facilities and the availability of advanced cardiac life support including induced hypothermia should be predetermined [24].

Practice and Review of the EAP

Each EAP should be regularly rehearsed and reviewed at least once annually by all persons responsible for the care of an athlete during an emergency. Ideally this occurs prior to the first practice of the season with local EMS personnel present to foster communication and awareness of a comprehensive plan. Additional training should occur when new staff members who may be involved in an emergency response are hired.

A "medical timeout" is encouraged prior to each sporting event with review of the EAP with the visiting team medical and/or coaching staff. These event-specific reviews should designate responder roles and account for personnel changes and construction or other barriers in the surrounding environment.

Post-Incident Review

After any SCA incident, a debriefing should occur. The event and emergency response should be reviewed and critically assessed. All equipment used should be returned to working order including any service or part replacement necessary. Psychological support for responders and bystanders should be arranged as deemed necessary on an individual or group basis.

Atypical Sporting Equipment Considerations

An American football or hockey athlete wearing full pads and a helmet who suffers SCA presents additional challenges for an effective resuscitation. If an athlete is suspected to be in SCA, opening or removal of the shoulder pads and helmet to allow access to the chest and airway and facilitate a rapid resuscitation is the priority. The following steps should be performed:

1. Cut the jersey and undershirt in a "T" fashion (sleeve to sleeve and down the middle from collar to waist) to expose the chest for CPR and AED lead placement.
2. Cut the shoulder pad straps to allow additional access to the chest and prepare for shoulder pad removal later.
3. Remove the face mask to access the airway; alternatively, the helmet chin strap can be cut to facilitate helmet removal.
4. When enough rescuers are present to assist, the athlete can be lifted and the shoulder pads removed by dragging flat along the ground from the head of the athlete. (Note that in a witnessed arrest, provision of chest compressions and application of the AED leads is the priority and should not be delayed to fully remove the shoulder pads and helmet unless this is necessary to facilitate access to the chest and/or airway.)

Additional sports or positions within a sport such as lacrosse or a baseball/softball catcher may have similar equipment which limits access to the thorax. The development of any EAP should directly evaluate and address the equipment worn by athletes and personnel trained accordingly.

Fig. 2 Approach to rowing shell for a coaching launch with an extended pontoon bow. The boats should be brought parallel to each other with the extracting boat on the non-oar side of the collapsed rower

Atypical Sporting Venue Considerations

While most sporting events take place in a single, confined space (e.g., a stadium), several sports involve competitions that involve a large geographical area. These include rowing, cycling, open water swimming, Nordic and downhill skiing, surfing, and distance running events. These events pose unique challenges for the development of an EAP.

Marathon Event Coverage

Long-distance running and other mass participation events occur frequently across the globe. Emergency medical care is ideally provided at regular intervals (i.e., every 1–2 km) across the length of the event course. Mobile rescuers on bicycles who are trained to recognize SCA, who can activate the EAP promptly, and who are equipped with an AED have demonstrated improved survival during marathons [20]. SCA is more likely to occur late in races and AED and emergency personnel distribution should reflect this variability in risk [25, 26].

Rowing Event Coverage

Rowing is a sport which involves training in two distinct environments. The first, land-based training, involves use of stationary equipment (typically a rowing ergometer) in a confined space. The second, water-based training, involves rowing in shells containing 2 to 9 individuals and typically covers a large geographical region with unpredictable conditions and risks (i.e., weather conditions, other water craft, variable support staff, and coaching). Development of an EAP for each of these training situations is critical. The land-based training location EAP should follow the principles outlined above for most other sporting venues. Water-based training EAPs need further thought and development to address issues unique to responding to an SCA while on the water.

The first unique challenge to consider is the extraction of a collapsed athlete from the rowing shell. Rowing shells will typically contain other individuals whose safety must also be considered. Coaches and support staff may be in the general vicinity of the rowing shells but the style of boat (coaching launch) they are in may vary.

Fig. 3 Approach to a rowing shell for a coaching launch with a flat deck launch. The boats should be brought perpendicular to each other with the bow of the extracting boat on the non-oar side of the collapsed rower

Fig. 4 Use of a simple nylon rescue strap, looped around the collapsed rower, greatly improves the ability of the rescuers to extract the collapsed rower from the rowing shell, prevent water immersion, and can help address discrepancies in body size between the athlete and rescuers

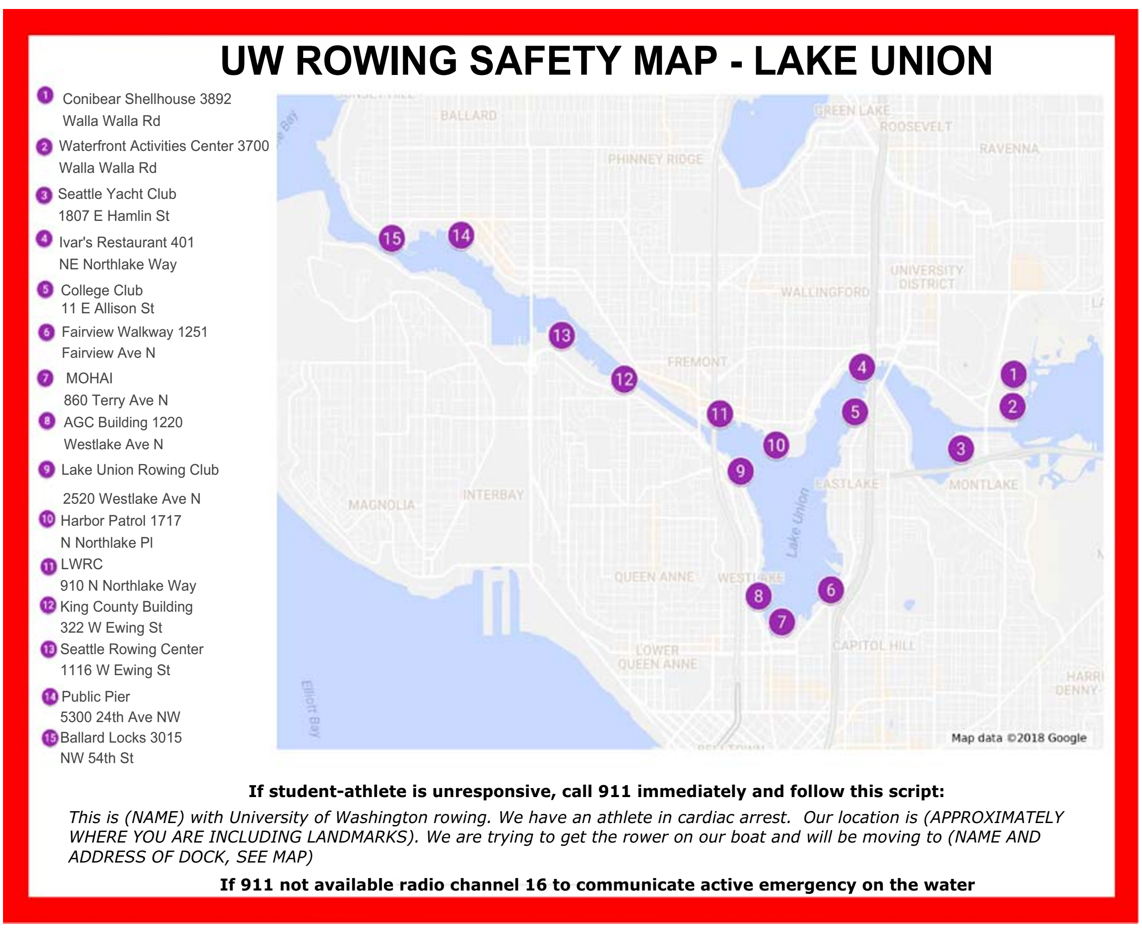

Fig. 5 Example emergency dock location map for a waterway used for rowing training including the address for each dock and script to follow when calling EMS. This map is posted in each coaching launch and rowing shell

The recommended technique for extracting a collapsed athlete from a rowing shell is shown in Figs. 2, 3, and 4. A rescue strap is placed around the torso and beneath the arms to assist transfer of the athlete from the shell to the launch (or rescue craft).

A second challenge faced in the context of a collapsed rower on the water is delivery to emergency medical personnel. On water rowing training often occurs several kilometers and considerable travel time away from primary, established EAP locations. It is important to work with local EMS to determine the optimal method and location to transfer a collapsed athlete to the next appropriate level of medical care. Extraction dock locations spread throughout the water training region that are accessible from both land and water for coach and emergency medical personnel should be identified. Specific location addresses should be listed on a map which is placed in all coaching launches and carried by all coxswains in a rowing shell (Fig. 5). When the EMS system is activated, the caller informs EMS of their current approximate location, based on landmarks, and the address of the dock to which the collapsed athlete is being taken. Land-based EMS services will then be directed to the address of the target dock. If water-based EMS is able to intercept the coaching launch on this path, the collapsed individual is transferred for faster transport.

The principles of applying an AED, starting and continuing CPR, and communicating with emergency medical personnel should follow the protocols outlined above.

For further information and discussion regarding a rowing specific, on-water, EAP please visit www.wateremergencytraining.com.

Conclusion

An EAP for SCA includes rescuers trained in CPR and the recognition of SCA, access to an AED, all staff awareness of the location of AEDs, and a means to activate the EMS system. AEDs should be located on-site at athletic venues or strategically placed so they can be retrieved and used within 3 min of an athlete's collapse. The EAP for SCA should be reviewed and rehearsed at least once annually by all individuals responsible for the care of an athlete during an emergency. A "medical timeout" should be completed before each sporting event begins and should include a readiness check of the AED, identification of key roles and personnel, communication methods, and venue-specific EAP information. Careful attention should be paid to the unique challenges of each sport and venue being covered including sport-specific equipment as well as access to and distance from emergency medical services.

Compliance with Ethical Standards

Conflict of Interest Henry F Pelto declares that he has no conflict of interest.

Jonathan A Drezner declares that he has no conflict of interest.

Ethical Approval This article does not contain any studies with human participants or animals performed by any of the authors.

References

1. Harmon, K. G., Asif, I. M., Maleszewski, J. J., et al. (2016). Incidence and etiology of sudden cardiac arrest and death in high school athletes in the United States. *Mayo Clinic Proceedings, 91*(11), 1493–1502.
2. Maron, B. J., Doerer, J. J., Haas, T. S., Tierney, D. M., & Mueller, F. O. (2009). Sudden deaths in young competitive athletes: Analysis of 1866 deaths in the United States, 1980-2006. *Circulation., 119*(8), 1085–1092.
3. Harmon, K. G., Asif, I. M., Maleszewski, J. J., et al. (2015). Incidence, cause, and comparative frequency of sudden cardiac death in National Collegiate Athletic Association Athletes: A decade in review. *Circulation., 132*(1), 10–19.
4. Maron, B. J., Haas, T. S., Murphy, C. J., Ahluwalia, A., & Rutten-Ramos, S. (2014). Incidence and causes of sudden death in U.S. college athletes. *Journal of the American College of Cardiology, 63*(16), 1636–1643.
5. Toresdahl, B. G., Rao, A. L., Harmon, K. G., & Drezner, J. A. (2014). Incidence of sudden cardiac arrest in high school student athletes on school campus. *Heart Rhythm, 11*(7), 1190–1194.
6. Drezner, J. A., O'Connor, F. G., Harmon, K. G., et al. (2016). AMSSM position statement on cardiovascular Preparticipation screening in athletes: Current evidence, knowledge gaps, recommendations, and future directions. *Clinical Journal of Sport Medicine, 26*(5), 347–361.
7. Hainline, B., Drezner, J. A., Baggish, A., et al. (2016). Interassociation consensus statement on cardiovascular care of college student-athletes. *Journal of the American College of Cardiology, 67*(25), 2981–2995.
8. Drezner, J. A., & Rogers, K. J. (2006). Sudden cardiac arrest in intercollegiate athletes: Detailed analysis and outcomes of resuscitation in nine cases. *Heart Rhythm, 3*(7), 755–759.
9. Drezner, J. A., Rao, A. L., Heistand, J., Bloomingdale, M. K., & Harmon, K. G. (2009). Effectiveness of emergency response planning for sudden cardiac arrest in United States high schools with automated external defibrillators. *Circulation., 120*(6), 518–525.
10. Corrado, D., Basso, C., Rizzoli, G., Schiavon, M., & Thiene, G. (2003). Does sports activity enhance the risk of sudden death in adolescents and young adults? *Journal of the American College of Cardiology, 42*(11), 1959–1963.
11. Meeuwisse, W. H., Schneider, K. J., Dvořák, J., et al. (2017). The Berlin 2016 process: A summary of methodology for the 5th international consensus conference on concussion in sport. *British Journal of Sports Medicine, 51*(11), 873–876.
12. Drezner, J. A. (2009). Preparing for sudden cardiac arrest–the essential role of automated external defibrillators in athletic medicine: A critical review. *British Journal of Sports Medicine, 43*(9), 702–707.
13. Neumar, R. W., Shuster, M., Callaway, C. W., et al. (2015). Part 1: Executive summary: 2015 American Heart Association guidelines update for cardiopulmonary resuscitation and emergency cardiovascular care. *Circulation., 132*(18 Suppl 2), S315–S367.

14. Monsieurs, K. G., Nolan, J. P., Bossaert, L. L., et al. (2015). European resuscitation council guidelines for resuscitation 2015: Section 1. Executive summary. *Resuscitation., 95*, 1–80.
15. Sasson, C., Rogers, M. A., Dahl, J., & Kellermann, A. L. (2010). Predictors of survival from out-of-hospital cardiac arrest: A systematic review and meta-analysis. *Circulation. Cardiovascular Quality and Outcomes, 3*(1), 63–81.
16. Bækgaard, J. S., Viereck, S., Møller, T. P., Ersbøll, A. K., Lippert, F., & Folke, F. (2017). The effects of public access defibrillation on survival after out-of-hospital cardiac arrest: A systematic review of observational studies. *Circulation., 136*(10), 954–965.
17. Blom, M. T., Beesems, S. G., Homma, P. C., et al. (2014). Improved survival after out-of-hospital cardiac arrest and use of automated external defibrillators. *Circulation., 130*(21), 1868–1875.
18. Berdowski, J., Schulten, R. J., Tijssen, J. G., van Alem, A. P., & Koster, R. W. (2010). Delaying a shock after takeover from the automated external defibrillator by paramedics is associated with decreased survival. *Resuscitation., 81*(3), 287–292.
19. Drezner, J. A., Toresdahl, B. G., Rao, A. L., Huszti, E., & Harmon, K. G. (2013). Outcomes from sudden cardiac arrest in US high schools: A 2-year prospective study from the National Registry for AED use in sports. *British Journal of Sports Medicine, 47*(18), 1179–1183.
20. Kinoshi, T., Tanaka, S., Sagisaka, R., et al. (2018). Mobile automated external defibrillator response system during road races. *The New England Journal of Medicine, 379*(5), 488–489.
21. Link, M. S., Myerburg, R. J., Estes, N. A., & American Heart Association Electrocardiography and Arrhythmias Committee of Council on Clinical Cardiology CoCDiY, Council on Cardiovascular and Stroke Nursing, C.uncil on Functional Genomics and Translational Biology, and American College of Cardiology. (2015). Eligibility and Disqualification Recommendations for Competitive Athletes With Cardiovascular Abnormalities: Task Force 12: Emergency Action Plans, Resuscitation, Cardiopulmonary Resuscitation, and Automated External Defibrillators: A Scientific Statement From the American Heart Association and American College of Cardiology. *Circulation, 132*(22), e334–e338.
22. Drezner, J. A., Courson, R. W., Roberts, W. O., Mosesso, V. N., Link, M. S., & Maron, B. J. (2007). Inter-association task force recommendations on emergency preparedness and management of sudden cardiac arrest in high school and college athletic programs: A consensus statement. *Clinical Journal of Sport Medicine, 17*(2), 87–103.
23. Casa, D. J., Almquist, J., Anderson, S. A., et al. (2013). The inter-association task force for preventing sudden death in secondary school athletics programs: Best-practices recommendations. *Journal of Athletic Training, 48*(4), 546–553.
24. Bernard, S. A., Gray, T. W., Buist, M. D., et al. (2002). Treatment of comatose survivors of out-of-hospital cardiac arrest with induced hypothermia. *The New England Journal of Medicine, 346*(8), 557–563.
25. Webner, D., DuPrey, K. M., Drezner, J. A., Cronholm, P., & Roberts, W. O. (2012). Sudden cardiac arrest and death in United States marathons. *Medicine and Science in Sports and Exercise, 44*(10), 1843–1845.
26. Redelmeier, D. A., & Greenwald, J. A. (2007). Competing risks of mortality with marathons: Retrospective analysis. *BMJ., 335*(7633), 1275–1277.